

3D Coordinates · Recreate the Statues (x, y, z) — English Worksheet (Advanced)

Topic: Recreate the Statues (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

A statue is many **stacked fill boxes**: legs, a torso, arms, a head. One **fill** makes a solid box between **two corners** (x, y, z) — x **across**, y **up**, z **forward**, feet on the **ground** (y = 0).

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict

Read each box. Write what you will see, **where it sits, and the reason**.

```
fill GRAY from (3, 0, 4) to (3, 2, 4)
fill GRAY from (5, 0, 4) to (5, 2, 4)
```

Two thin boxes, x = 3 and x = 5, both y = 0–2. What part is this, why two of them, and why y = 0?


```
fill GRAY from (2, 3, 4) to (2, 5, 4)
fill GRAY from (6, 3, 4) to (6, 5, 4)
```

These start at y = 3 and sit at x = 2 and x = 6 — outside the legs. What part is this, and why is the x outside?


```
fill WHITE from (3, 7, 4) to (5, 8, 4)
```

This is the highest box ($y = 7-8$). What part is it, and how do you know from the y?


```
fill GRAY from (3, 3, 4) to (5, 6, 4)
fill WHITE from (3, 7, 4) to (5, 8, 4)
```

Two boxes together. The torso ends at $y = 6$ but the head starts at $y = 7$. What sits between them, and what does that gap mean?

2 · Spot the Bug 🐛

Each block is broken. Read what it should do, write the fixed code, then explain.

Bug A — The torso should sit **on top** of the legs. The legs go $y = 0-2$, so the torso should start at $y = 3$. Right now it starts at $y = 0$, so nothing rises.

```
fill GRAY from (3, 0, 4) to (5, 6, 4)
```

Hint: the **2nd number (y)** is height. Where should the torso's bottom y be?

Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

Bug B — This should be the **right arm**, hanging on the **side** of the torso at $x = 6$. The torso fills $x = 3-5$, so $x = 4$ is **inside** it — the arm disappears into the body.

```
fill GRAY from (4, 3, 4) to (4, 5, 4)
```

Hint: to be on the outside, the arm's **1st number (x)** must be past the torso's $x = 3-5$.

Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

Bug C (*the tricky one*) — Your friend's whole statue is **floating** — a gap of air under the feet. The legs are built at **y = 1–3** instead of touching the ground. The feet must sit on the floor.

```
fill GRAY from (3, 1, 4) to (3, 3, 4)
fill GRAY from (5, 1, 4) to (5, 3, 4)
```

Hint: the ground is **y = 0**. What is the lowest y a foot should have?

Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

3 · Show Your Work 📷 🎥

Walk around a statue in your world. Then plan and build a **full golem** on your home spot — **6 fill boxes**: two legs, a torso, two arms, and a head.

📊 Recipe (every corner is **(x, y, z)**, feet on the ground $y = 0$): - left leg: gray from (3, 0, 4) to (3, 2, 4) - right leg: gray from (5, 0, 4) to (5, 2, 4) - torso: gray from (3, 3, 4) to (5, 6, 4) - left arm: gray from (2, 3, 4) to (2, 5, 4) - right arm: gray from (6, 3, 4) to (6, 5, 4) - head: white from (3, 7, 4) to (5, 8, 4)

Before you build, **plan the heights** in the space below: which y does each part start and stop at?

🌟 **Modify challenge:** after it works, change one thing of your own — a new colour, a different pose, or an extra part (a hat, a tail, longer arms). Write what you changed.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.